Image Based Extraction of Heart Rate, Blood Pressure and Perfusion Using Artificial Intelligence

Garrick Jensen[1] , Justin Peel PhD[1], Kelly Roman[1], Nathan Fisher[2], Ethan Quist[2]

[1]Areté, Northridge, CA, [2]Telemedicine and Advanced Technology Research Center, Fort Detrick, MD



Objectives

- Demonstrate a small transportable contactless vitals monitoring system
- Leverage machine learning and advanced signal processing to declutter vitals info present in video data

Background

- Current vitals monitoring requires a host of contact sensors that requires time and attention to apply and utilize
- Our solution is fully remote and accurately records a host of vitals measurements with only video data
- Previous attempts at automated vitals recognition are often limited by noise and non-ideal conditions
- With custom deep neural networks, regions with high Signal to Noise Ratio (SNR) can be identified to remove clutter and operate even with lighting and motion artifacts
- We have created a non-contact vitals monitor that is easily transported and can read vitals data in confined spaces or from a drone



Multi-Camera Collection System: *4 high resolution, low-light cameras, designed to collect high quality data for development. System mounts to basket or mast.*

Single Camera Collection System: *1 camera, designed primarily for in field data collections. Mounted to tripod for easy mobility.*

Drone Mounted Collection System: *Single camera mounted on drone, designed for covering large areas and identifying patients in difficult terrain.*

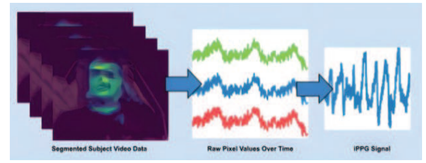


Stacked Litter Use-Case: *Multiple cameras allow for remote processing on the entire subject even in small spaces*

Methods

Human and Skin Segmentation:

- Measurements are derived from video using several layered algorithms
- Subjects are identified using a deep learned body pose and key points detection algorithms
  - We not only identify the person, but segment the subjects into 6 distinct limbs and 17 keys points
- Our skin segmentation network identifies regions of our subject that give us the best SNR
- Our custom body and skin networks are compact to run in real time and trained on in-house subject data



iPPG Signal Processing Data Flow



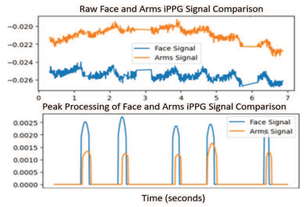
Custom CNNs segment areas with PPG signal for finer signal extraction

Non-Contact Heart Rate:

- Heart rate is measured through changes in skin tone, with a technique called iPPG
- Skin segmentation is combined with body segmentation to ensure we extract only high SNR regions
- The full PPG waveform is measured on all exposed skin and can be divided by limb location
- We can analyze the perfusion of each limb and find body parts with stronger signal

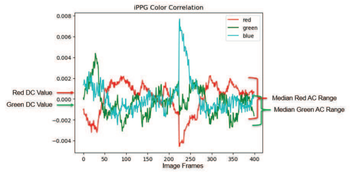
Non-Contact Blood Pressure:

- Pulse Transit Time (PTT) is the measured time for the pulse to travel from the heart to another location
- PTT is directly correlated with blood pressure
- Our system plans to monitor changes in blood pressure via PTT, but not absolute pressure



Face and Arm iPPG Comparison of Raw and Processed Data

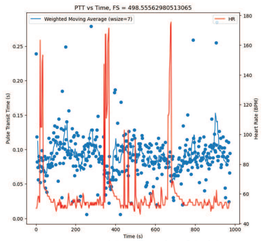
- PTT is subject dependent, measuring changes in blood pressure rather than absolute values saves on time consuming calibration
- The time delay between the pulse arrival in the face and arms is measured



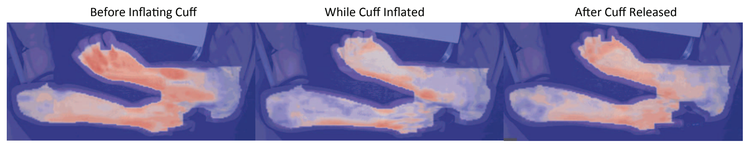
- Strong negative red and green channel correlation implies color change due to perfusion rather than lighting
- The ratio between the red and green AC and DC values indicate how much of each frequency is absorbed
- The AC and DC ratio changes when blood oxygenation changes

Non-Contact Limb Perfusion:

- Perfusion is a measure of blood flow within the limb
- Tourniquets and other obstructing devices affect perfusion and circulation
- Perfusion is measured with an optical camera by measuring:
  - green and red channel correlation
  - Ratio of AC and DC values of iPPG signal



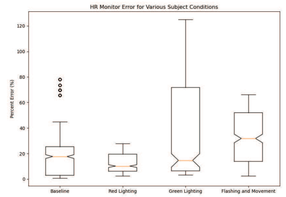
Heart Rate and PTT Study of Subjects undergoing Valsalva Maneuver



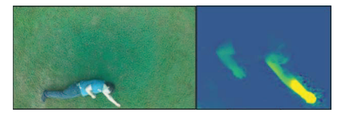
Measuring Perfusion on Patient with Blood Pressure Cuff Applied to Arm

Results

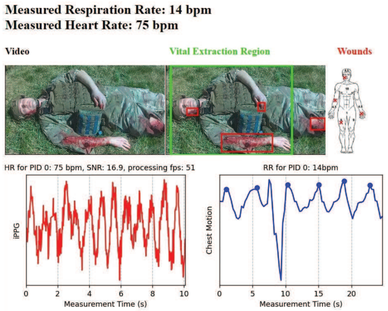
- We have successfully created algorithm that robustly measure heart rate and detect changes in blood pressure and perfusion
- We found strong agreement between our measurements and the ground truth with HR measurements within 5 BPM of ground truth
- We have tested our blood pressure and perfusion monitors on subjects under-going physiological stress and have had promising results



Heart Rate Vitals Accuracy in Non-ideal Conditions



Skin Segmentation on Subject from Drone Imaging



Example GUI of Remote Vitals Monitoring Running in Real Time

Conclusion

We have illustrated non-contact vitals monitoring, extracting not only heart rate but also a blood pressure analoge and localized perfusion measurements. We have developed custom deep learned neural networks to identify regions of high SNR. In addition to our advanced signal processing, we can extract vitals in difficult movement and lighting conditions. We hope to fully incorporate pulse transit time for blood pressure monitoring along with perfusion monitoring onto the vitals monitoring system.

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